Conversion of Camphene to Camphor: Day 3

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CHEM 2081-003

Introduction

This experiment details the process of the conversion of borneol to camphor with glacial acetic acid, acetone, sodium hypochlorite, and DCM. This process was achieved by a gentle reflux, followed by extraction. The final product was then dried with anhydrous sodium sulfate and heated over a sand bath until solid formed.

Materials and Methods

To begin, the sand bath was plugged into the 10 Amp outlet and set to power 60, and a 100 mL beaker filled halfway with water was placed in the bath. Next, 0.3 g of borneol and 0.3 mL of acetic acid were added in a long neck round bottom flask. The mixture was then dissolved in 0.3 mL of glacial acetic acid, and then a boiling chip was placed into the flask. Following, a reflux apparatus was assembled. After the borneol was dissolved completely, the long neck round bottom flask was placed in the water bath, where the temperature of the water stayed in the 40℃-50℃ range. Next, 6 mL of aqueous sodium hypochlorite was added to the flask in a drop-wise manner over 35 minutes. After each addition of sodium hypochlorite, the mixture was gently swirled. Once all the sodium hypochlorite was added to the flask, the reaction was allowed to reflux for an additional 15 minutes. Following, the mixture was allowed to cool to room temperature and then was transferred to a centrifuge tube. The camphor was extracted twice using 2 mL portions of DCM each time. To neutralize the combined DCM layers, 2 mL of saturated sodium bicarbonate was added to the centrifuge. Once there were no more bubbles formed after the addition of the sodium bicarbonate, the top layer in the centrifuge was removed. The remaining organic layer was then washed with 2 mL of sodium bisulfite. The sodium bisulfite was removed from the centrifuge, and then the organic layer was washed again with 2

mL of water. After the water was removed, the organic layer was then dried with anhydrous sodium sulfate. Lastly, the liquid organic layer was decanted onto a watch glass and heated over the water bath to evaporate the DCM, leaving solid camphor, which was then weighed.

Results

Amount of Borneol used (g)	0.307
Amount of glacial acetic acid used (mL)	0.3
Amount of acetone used (mL)	1.0
Amount of aqueous sodium hypochlorite used (mL)	6.0
Amount of DCM used (mL)	4.0
Amount of saturated sodium bicarbonate (mL)	2.0
Amount sodium bisulfate (mL)	2.0
Amount of water (mL)	2.0

Table 1. Amount of chemicals used in the experiment.

Molar mass of Borneol (g/mol)	154.25
Molar mass of glacial acetic acid (g/mol)	60.052
Molar mass of acetone (g/mol)	58.08
Molar mass of sodium hypochlorite (g/mol)	74.44
Molar mass of camphor (g/mol)	152.23
Density of DCM (g/mL)	1.325
Density of sodium bicarbonate (g/mL)	2.16
Density of sodium bisulfate (g/mL)	1.48

Table 2. Molar mass values for the reactants and product, and densities of the chemicals used during the extraction process.

Theoretical yield (g)	0.303
Weight of watch glass (g)	33.072
Weight of watch glass and product (g)	33.296
Amount of Camphor collected (g)	0.224

Table 3. Theoretical yield and actual yield of camphor.

Calculations

Limiting reagent calculations

$$0.307\ g\ Borneol\ \times\ (1\ mol\ \div\ 154.25\ g)\ \times\ (1\ mol\ Camphor\ \div\ 1\ mol\ Borneol)\ =\ 0.0019903$$

theoretical moles of Camphor

Percent yield calculations

$$0.0019903\ mol\ Camphor\ \times\ (152.23\ g\ \div\ 1\ mol)\ =\ 0.303$$ *theoretical grams of Camphor*

$$33.296\ g\ -\ 33.072\ g\ =\ 0.224$$ *grams of Camphor collected*

$$(0.224\ g\ \div\ 0.303\ g)\ \times\ 100\%\ =\ 73.9\%$$ *yield*

Discussion

In this experiment, 0.224 g of camphor was produced by refluxing the reaction and extracting the final product. The theoretical yield of this experiment was 0.303 g of camphor, resulting in a 73.9% yield. Possible sources of error include that the starting material of borneol could have been impure, that the reaction reflux was out of the appropriate temperature range, or that the addition of aqueous sodium hypochlorite was too fast. Because of the amount of camphor collected, and confirmed by the thermometer reading, it can be deduced that the heating

during reflux was at a higher than preferred temperature. The reaction is already exothermic, and the thermometer would read as high as 65℃ over the 35 minute reflux. One way to avoid this error would be to monitor and adjust the temperature of the sand bath more precisely, and to ensure that the sodium hypochlorite is added at a slow rate and that the round bottom flask is agitated after each drop.

References

1. H McNeil. *Chem 2081 Organic Chemistry II Lab Manual*. Macmillan Learning Curriculum Solution: Plymouth, 2023, (p. 65-66)

Lab Performance

Exp 6 D3

Name: Lauren Kuykendall Section: 003 Room: 402 Hood:

Wearing personal protection equipment—this evaluation records your consistency with wearing the appropriate clothes for lab (long pants, closed toe shoes, and tops with sleeves) as well as the required laboratory safety equipment of apron, goggles, and gloves.	○ ○ ○ ○ ○ (5pts)
Safe working habits—as the GTAs do the lab rotations, they will record your work ethic and whether you follow our safety rules and regulations concerning tied hair, tied apron, safety goggles on at all times, and no food and drinks.	○ ○ ○ ○ ○ (5pts)
Following our laboratory safety technique work procedures—evaluation of operation of our laboratory equipment—water sinks in hood (1pt), vacuum and air lines (1pt), sash position (1pt), and heating elements (1pt). It is important that after every experiment you shut down the hood and make it ready for the section after yours (5pts).	○(1pt) ○(1pt) ○(1pt) ○ (1pt) ○ (5pts)
Glassware operation—safe handling of our glassware equipment. This means cleaning (1pt), maintaining the drawer inventory (5pts), and proper experimental setup (5pts).	○(1pt) ○ (5pts) ○ (5pts)
Working independently or equal partner participation during the entire lab.	○ ○ ○ ○ ○ (5pts)
Good work behavior—you work in a dynamic environment; therefore, creating distractions through noise and disobedience will not be tolerated. No cell phone use, listening to loud music, or working/distracting at a non-assigned workspace.	○ ○ ○ ○ ○ (5pts)
Maintain clean and organized personal space.	○ ○ ○ ○ ○ (5pts)

GTA notes:

GTA signature: Final Performance Grade: 8

Exp. No.	Experiment/Subject	Date	
6 day 3	Conversion of camphene to camphor		
Name	Lab Partner	Locker/ Desk No.	Course & Section No.
lauren kuykendall			

Materials

sand bath
long neck roundbottom flask
centrifuge
boiling chip
reflux set up
water bath
pipette

0.3g borneol
0.3 mL glacial acetic acid
1 mL acetone
6 mL aq sodium hypochloride
4 mL DCM
2 mL saturated sodium bicaroonate
2 mL sodium bisulfite
2 mL water
anhydrous sodium sulfate

Procedures

1. In the long neck round bottom flask, add racemic 0.3g of borneol with 0.3mL glacial acetic acid. Dissolve the mixture in 1 mL acetone. Add a boiling chip.
2. Assemble a reflux with the distillation column & place the apparatus in a water bath with temp. between 40°C-50°C. If the borneol doesn't dissolve, add 0.5 mL of acetone until all borneol goes into solution.
3. Once dissolved, add 6mL of aq. sodium hypochloride solution. The addition should be done through the top opening of the condenser & should be paced dropwise over 35 minutes. After each addition of the hypochloride, the mixture should be given a gentle swirl. The reaction must be allowed to continue for 15 minutes after the last hypochloride addition.
4. Next, let the mixture cool to room temperature & extract the camphor with two portions (2mL each) of DCM. To make the extraction more efficient, use a centrifuge tube.
5. Neutralize the combined methylene chloride layers with ~2mL of saturated sodium bicarbonate. Keep adding bicarbonate solution if necessary. The presence of acidic residuals will be determined by the production of CO_2 bubbles after every addition of the bicarbonate solution.
6. Next, wash the organic solution sequentially with 2mL of 5% sodium bisulfite & 2mL of water
7. Dry the organic solution over anhydrous sodium sulfate. The solution is considered dry when freely rolling crystals of the drying agent are observed.
8. Evaporate the DCM over a water bath with temp not exceeding 50°C. Weigh the collected crystals & determine % yield. Compare melting points

Date	Witness/TA	Date